

Business Analysis Portfolio Case Study

Market Analysis & UI Audit: Streamlining E-Commerce Checkout

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1. Executive Summary

In modern e-commerce systems, user retention during the final stages of the purchase funnel is critical for maximizing sales conversion rates. This case study focuses on analyzing current customer onboarding flows in major e-commerce applications and optimizing the checkout phase. By diagnosing high-friction touchpoints, this project delivers a business requirement and visual workflow to replace traditional, extensive authentication barriers with a low-friction, passwordless OTP (One-Time Password) system combined with an optimized "Guest Checkout" option.

2. Problem Statement & Market Analysis

During the competitive benchmarking and UI audit phase of prominent e-commerce portals (such as Amazon and Myntra), a major point of **User Friction** was identified during the transition from product selection to purchase finalization.

- **Mandatory Registration Obstacles:** Forcing non-logged-in or unregistered users to complete comprehensive multi-field registration forms (demanding full name, verified email, password creation, etc.) creates immediate irritation.
- **Cart Abandonment (Drop-off Rate):** Lengthy authentication flows cause significant cognitive overload. Users looking for a rapid purchase experience frequently close the application at this junction, leading to elevated cart abandonment rates and direct revenue leakage for the business.

3. Proposed Solution & Value Proposition

To neutralize user irritation and lower the drop-off rates, a streamlined checkout architecture is introduced, featuring two main components:

1. **Passwordless Quick Checkout:** Users authenticate seamlessly using just their localized 10-digit mobile number and a rapid 4-digit SMS OTP, totally eliminating email verification and password creation steps from the critical path.
2. **Gated Pricing Strategy:** To drive user registration without creating forced friction, a dual-pricing model is applied. All users view standard pricing and regular discounts. However, an incremental incentive tag (e.g., *"Login to unlock an extra 10% Member Discount"*) is positioned clearly, naturally motivating high-value users to complete the quick-login process willingly.

4. Agile Product Requirement (User Story)

To translate this business requirement into an actionable item for technical development teams, the feature is mapped into a standard Agile User Story format:

- **AS AN** E-commerce customer/buyer,

- **I WANT TO** seamlessly login or check out using my mobile number and a 4-digit OTP instead of traditional emails and passwords,
- **SO THAT** I can complete my purchase quickly, securely, and without any registration-induced irritation.

5. Functional Specifications & Acceptance Criteria

Acceptance Criteria (AC) define the functional boundaries and exact rules that the technical development and quality assurance teams must meet.

- **AC-01 (Country Code Localization):** The system must provide an interactive dropdown menu to let the user select their respective country code prior to inputting their contact number.
- **AC-02 (Input Field Validation):** The mobile number input field must strictly accept numerical values and validate that the entry conforms to exactly a 10-digit format.
- **AC-03 (OTP Generation & Format):** Upon clicking the "Send OTP" trigger, the system must generate and dispatch a secure, 4-digit numeric verification code to the designated device.
- **AC-04 (OTP Expiration & Retry Loop):** The generated OTP must expire automatically after exactly 60 seconds. Upon expiration, a "Resend OTP" link must become active.
- **AC-05 (Error Feedback Handling):** If a user inputs an invalid or expired OTP, the system must flag a clear warning message ("Enter valid OTP") and allow immediate re-entry.

6. Process Workflow Mapping (Logical Flowchart Description)

The backend systems and functional user interface interactions follow a rigorous structural flow:

1. **[Step 1]** User selects an item and clicks on "Login/Checkout".
2. **[Step 2]** System displays the unified modal asking for Country Code and a 10-digit Mobile Number.
3. **[Step 3]** System processes input, generates a 4-digit numeric code, and dispatches it via SMS gateway.
4. **[Step 4]** User encounters the verification prompt and inputs the received 4-digit OTP.
5. **[Step 5 - Decision Gate]** System runs verification: Is the OTP match correct AND completed within 60 seconds?
 - **Branch NO:** System renders the inline error message "Enter valid OTP" and routes the user back to step 4 for a retry loop.
 - **Branch YES:** System completes secure authentication, bypasses registration pages, and directly advances the user to the Delivery Address and Payment Page.

7. Key Metrics for Success (KPIs)

To measure the practical business impact post-deployment, the Business Analyst tracks the following Key Performance Indicators:

- **Checkout Conversion Rate:** Expected upward trend in finalized transactions.
- **Cart Abandonment Rate:** Targeted reduction in user drop-offs directly at the sign-in step.

- **Average Time-to-Checkout:** Decreased total time duration taken by a non-logged-in user to complete an order.